SDS 8102: Advanced Mathematical Statistics

Midterm Examination

Fall 2026

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Name: _____ **Student ID:** _____

Date: *Oct 14, 2026* **Duration:** *1 hr 30 mins.*

Instructions

1. This examination contains **four questions**. Answer all questions.
2. The examination is graded out of **50 points**. No buffer points.
3. The midterm contributes **35%** of the final course grade.
4. Explain your reasoning as clearly and completely as possible. Credit will be awarded for clear mathematical explanations, even when the final answer is incomplete.
5. Clearly state every theorem or result that you use, and explain why its assumptions are satisfied in the setting of the problem. Merely naming a theorem is not sufficient.
6. Show all important intermediate steps. A correct final answer without adequate justification may receive only partial credit.
7. Mathematically correct alternative arguments are welcome and will receive full credit. If a minor computational error occurs after a correct setup, partial credit may still be awarded.
8. If you are unable to complete one part of a question, you may use the stated conclusion of that part in subsequent parts.
9. Unless otherwise stated, all random variables are defined on a common probability space, and all expectations and probabilities are assumed to be well-defined.

Write your name and student ID on every answer booklet. If you require additional space, clearly indicate where the continuation of your solution appears.

Question	1	2	3	4	Total
Available	12	12	10	16	50
Score					

Question 1 Poisson distribution and UMVUE (12 Points)

Let $n \in \mathbb{N}$ and $X_1, X_2, , \ldots, X_n \overset{i.i.d.}{\sim} \text{Poisson}(\lambda)$ for an unknown parameter $\lambda > 0$.

(a) Find a complete sufficient statistic for λ. Prove both sufficiency and completeness. [5 points]

(b) Find the UMVUE of $\exp(-\lambda)$. Justify your answer fully. [4 points]

(c) Show that $\sqrt{\lambda}$ cannot be unbiasedly estimated. [3 points]

Solution:

(a) The joint distribution fo $X_1, \cdots, X_n$ is

$$f_\lambda(x_1, \ldots, x_n) = \underbrace{\frac{1}{\prod_{j=1}^n x_j!}}_{h(x_1,\ldots,x_n)} \times \underbrace{\exp\left\{ (\sum_{j=1}^n x_j) \log \lambda - n\lambda \right\}}_{g_\lambda(\sum_{j\in[n]} x_j)}.$$

Therefore, by the Nyeman-factorization lemma $T(\mathbf{X}) := \sum_{i=1}^n X_i$ is sufficient statistics and follows the Poisson$(n\lambda)$ distribution.

To show completeness, let $\mathbb{E}_\lambda[g(T)] = 0$ for a measurable $g(\cdot)$, i.e.,

$$\sum_{k=0}^{\infty} g(k) e^{-n\lambda} . \frac{(n\lambda)^k}{k!} = 0 \quad \text{for all } \lambda > 0.$$

The above is a power series that is identically 0 on the positive half-line. Therefore, all the coefficients must be 0, i.e., $g(k) = 0$ for all $k \in \mathbb{N} \cup \{0\}$. This shows that $T(\mathbf{X})$ is also complete.

(b) Note that $\mathbb{1}(X_1 = 0)$ is an unbiased estimator of $\exp(-\lambda)$. By Lehmann-Scheffe theorem, the unique UMVUE is

$$\begin{aligned}
\mathbb{P}(X_1 = 0 \mid T = t) &= \frac{\mathbb{P}(X_1 = 0, \sum_{i=2}^n X_i = t)}{\mathbb{P}(\sum_{i=1}^n X_i = t)} \\
&= \frac{e^{-\lambda} \times e^{-(n-1)\lambda} \times \frac{(n-1)^t \lambda^t}{t!}}{e^{-n\lambda} \frac{n^t \lambda^t}{t!}} \\
&= (1 - \frac{1}{n})^t.
\end{aligned}$$

Hence, the unique UMVUE of $\exp(-\lambda)$ is $(1 - \frac{1}{n})^T$.

(c) Suppose, $\sqrt{\lambda}$ is unbiasedly estimable by a L_1 estimator $U(\mathbf{X})$. Then, $\delta(T) := \mathbb{E}[U \mid T]$ is also an unbiased estimator of $\sqrt{\lambda}$. Since $T \sim \text{Poisson}(n\lambda)$, we arrive at

$$P(\lambda) := \mathbb{E}_\lambda[\delta(T)] = \sum_{k\geq 0} \delta(k) e^{-n\lambda} \frac{n^k \lambda^k}{k!} = \sqrt{\lambda} \quad \text{for all } \lambda > 0.$$

As $\lambda \mapsto \sqrt{\lambda}$ is differentiable, so is $P(\lambda)$. Differentiating both sides w.r.t λ we get $P'(\lambda) = \frac{1}{2\sqrt{\lambda}}$. This is contradiction as $P'(\lambda)$ is another power series that is continuous at $\lambda = 0$, and $\lambda \mapsto 1/(2\sqrt{\lambda})$ is not. This completes the proof.

Question 2 Gaussian Correlation and Sign Data (12 Points)

Let

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right), \qquad -1 < \rho < 1.$$

(a) A generative model $\mathcal{G}_\rho$ generates independent observations from the preceding bivariate Gaussian distribution. Based on m i.i.d samples from $\mathcal{G}_\rho$, construct a consistent estimator of ρ and find its asymptotic distribution. [5 points]

(b) Suppose, due to a corrupted output layer, the generator now returns only the sign values

$$\big(\text{sign}(X_i), \text{sign}(Y_i)\big), \qquad i = 1, \ldots, m,$$

rather than the values of X_i and Y_i. Using *only these sign observations*, construct a consistent estimator of ρ and find an asymptotic 95% confidence interval. [7 points]

Bonus: Which one is asymptotically more efficient? [No points]

Hint: It is a standard result that $\mathbb{P}_\rho(X > 0, Y > 0) = \frac{1}{4} + \frac{\arcsin(\rho)}{2\pi}$. You may use this result directly.

Solution:

(a) Note that $\mathbb{E}\,|XY| < \sqrt{\mathbb{E}(X^2)\mathbb{E}(Y^2)} < \infty$. Also, we have $\mathbb{E}(XY) = \text{Cov}(X, Y) = \rho$. Then, by WLLN, we can conclude that

$$T_m := \frac{1}{m}\sum_{i=1}^{m} X_i Y_i \xrightarrow{p} \rho.$$

To find the asymptotic distribution, we first note that

$$\begin{aligned} \mathsf{Var}(XY) &= \mathbb{E}[\mathsf{Var}(XY \mid X)] + \mathsf{Var}(\mathbb{E}[XY \mid X]) \\ &= \mathbb{E}[X^2 \mathsf{Var}(Y \mid X)] + \mathsf{Var}(X\mathbb{E}[Y \mid X]) \\ &= (1-\rho^2)\mathbb{E}[X^2] + \rho^2 \mathsf{Var}(X^2) \\ &= (1-\rho^2) + \rho^2 v, \qquad \text{where } v := \mathsf{Var}(X^2). \end{aligned}$$

Then, we can simply apply CLT to infer

$$\frac{\sqrt{m}(T_m - \rho)}{\sqrt{(1-\rho^2) + \rho^2 v}} \xrightarrow{d} \mathcal{N}(0, 1).$$

(b) From the given hint and the symmetry of the Guassian distribution, we first infer that

$$\mathbb{P}(\mathsf{sign}(X).\mathsf{sign}(Y) > 0) = 2\mathbb{P}(X > 0, Y > 0) = \frac{1}{2} + \frac{\arcsin \rho}{\pi}.$$

Write $U = \mathsf{sign}(X)$ and $V = \mathsf{sign}(Y)$ for notational brevity, and let us define

$$Q_m := \frac{1}{m}\sum_{i \in [m]} \left(\mathbb{1}\{U_i V_i > 0\} - \frac{1}{2}\right).$$

As the summands are i.i.d. and bouned, applying WLLN yields

$$Q_m \xrightarrow{p} \mathbb{P}(UV > 0) - 1/2 = \frac{\arcsin(\rho)}{\pi}.$$

Then, using *continuous mapping* theorem, we can conclude that

$$\widehat{\rho}_m := \sin(\pi Q_m) \xrightarrow{p} \rho \quad \text{as } m \to \infty.$$

For 95% CI, we start with the asymptotic distribution of Q_m. Write $\theta(\rho) := \frac{1}{2} + \frac{\arcsin(\rho)}{\pi}$, and apply CLT to get

$$\frac{\sqrt{m}\left(Q_m - \frac{\arcsin(\rho)}{\pi}\right)}{\sqrt{\theta(\rho)(1-\theta(\rho))}} \xrightarrow{d} \mathcal{N}(0,1).$$

Approach 1: The derivative of $x \mapsto \sin(\pi x)$ is $\pi\cos(\pi x)$. Therefore, applying the Delta theorem with the map $x \mapsto \sin(\pi x)$, an idealized 95% CI for ρ is

$$\left[\widehat{\rho}_m - \frac{z_{0.95}\pi^2(1-\rho^2)}{\sqrt{m}}\sqrt{\theta(\rho)(1-\theta(\rho))} \quad , \quad \widehat{\rho}_m + \frac{z_{0.95}\pi^2(1-\rho^2)}{\sqrt{m}}\sqrt{\theta(\rho)(1-\theta(\rho))}\right].$$

However, the deviance depends on the unknown parameter ρ. Therefore, due to continuous mapping theorem, we can construct the following data dependent 95% CI:

$$\left[\widehat{\rho}_m - \frac{z_{0.95}\pi^2(1-\widehat{\rho}_m^2)}{\sqrt{m}}\sqrt{\theta(\widehat{\rho}_m)(1-\theta(\widehat{\rho}_m))} \quad , \quad \widehat{\rho}_m + \frac{z_{0.95}\pi^2(1-\widehat{\rho}_m^2)}{\sqrt{m}}\sqrt{\theta(\widehat{\rho}_m)(1-\theta(\widehat{\rho}_m))}\right].$$

Approach 2 (alternative): An idealized 95% CI for $\arcsin(\rho)$ is

$$\left[\arcsin(\widehat{\rho}_m) \pm \frac{z_{0.95}\pi}{\sqrt{m}}\sqrt{\theta(\rho)(1-\theta(\rho))}\right].$$

It is easy to see that for large m, the interval completely lies inside $(-\pi/2, \pi/2)$ and $\sin(\cdot)$ is monotone there. Then, using continuous mapping theorem, a data-driven 95% CI is

$$\left[\sin\left(\arcsin(\widehat{\rho}_m) - \frac{z_{0.95}\pi}{\sqrt{m}}\sqrt{\theta(\widehat{\rho}_m)(1-\theta(\widehat{\rho}_m))}\right) \quad , \quad \sin\left(\arcsin(\widehat{\rho}_m) + \frac{z_{0.95}\pi}{\sqrt{m}}\sqrt{\theta(\widehat{\rho}_m)(1-\theta(\widehat{\rho}_m))}\right)\right]$$

Question 3 Alice's Personal-Record Problem (10 Points)

Alice has joined a gym and records her best weight lifted each week. Under a deliberately simple model, her weekly performances

$$X_1, X_2, \dots$$

are i.i.d. with a common continuous distribution function F. Alice declares week j to be a *personal-record (PR) week* if she has hit a new max-weight, i.e.,

$$X_j > \max\{X_1, \dots, X_{j-1}\}, \quad \text{for } j > 1.$$

For $j \geq 2$, define

$$I_j = \mathbf{1}\{X_j \text{ is a personal record}\}, \qquad R_N = \sum_{j=2}^{N} I_j,$$

so that R_N is the number of personal records achieved during the first N weeks.

(a) Show that $I_j \sim \text{Bernoulli}(\frac{1}{j})$. This shows that hitting new PR gets harder with time. [2 points]

(b) Prove that

$$\frac{R_N}{\log N} \longrightarrow 1 \qquad \textit{almost surely.}$$

[8 points]

Useful fact (hint). You may use the following version of **Kronecker's lemma**: if $(a_n)_{n\geq 1}$ is increasing with $a_n \to \infty$ and $\sum_{n=1}^{\infty} \frac{x_n}{a_n}$ converges, then $a_n^{-1} \sum_{k=1}^{n} x_k \longrightarrow 0$.

Solution:

(a) $X_1, X_2, \cdots$ are i.i.d. random variables. Hence, they are also exchangeable and their ranks can be uniformly arranged. If $I_j = 1$, then $X_1, \dots, X_{j-1}$ can be freely arranged in $(j-1)$ spots before X_j. Hence,

$$\mathbb{P}(X_j > \max\{X_1, \dots, X_{j-1}\}) = \frac{\#\text{ ways } X_1, \dots, X_{j-1} \text{ can be freely arranged before } X_j}{\#\text{ ways } X_1, \dots, X_j \text{ can be freely arranged}} = \frac{(j-1)!}{j!} = \frac{1}{j}.$$

This proves the claim.

(b) To prove independence, let R_j denote the rank of X_j among $X_1, \dots, X_j$. Since the distribution is continuous, there are no ties, and the relative ordering of $X_1, \dots, X_m$ is uniformly distributed over the $m!$ possible permutations.

The map from a permutation to its sequence of successive ranks

$$(R_1, \dots, R_m), \qquad R_j \in \{1, \dots, j\},$$

is a bijection. Therefore, for every admissible $(r_1, \dots, r_m)$,

$$\mathbb{P}(R_1 = r_1, \dots, R_m = r_m) = \frac{1}{m!} = \prod_{j=1}^{m} \frac{1}{j}.$$

Since

$$\mathbb{P}(R_j = r_j) = \frac{1}{j},$$

it follows that $R_1, \ldots, R_m$ are independent. Finally,

$$I_j = \mathbf{1}\{R_j = j\},$$

so $I_1, \ldots, I_m$ are independent. Since m is arbitrary, the entire sequence $\{I_j\}_{j \geq 1}$ is independent.

Now, we are ready prove the convergence. Define the independent rvs $Y_j = \frac{I_j - \frac{1}{j}}{\log j}$. We will check the Kolmogorov's condition:

$$\sum_{j=2}^{\infty} \mathsf{Var}(Y_j) = \sum_{j=2}^{\infty} \frac{\frac{1}{j}(1 - \frac{1}{j})}{\log^2 j} < \sum_{j=2}^{\infty} \frac{1}{j \log^2 j}$$

The last series converges as the tail sum

$$\sum_{j=N}^{\infty} \frac{1}{j \log^2 j} < \frac{1}{N \log^2 N} + \int_N^{\infty} \frac{dx}{x \log^2 x} = \frac{1}{N \log^2 N} + \frac{1}{\log N} \longrightarrow 0.$$

This shows that $\sum_{j=2}^{\infty} \frac{I_j - \frac{1}{j}}{\log j}$ converges almost surely. Now, we use Kronecker's lemma to conclude

$$\frac{R_N}{\log N} - \frac{\mathbb{E}[R_N]}{\log N} \longrightarrow 0 \quad \text{almost surely.}$$

Finally, note that $\mathbb{E}[R_N] = \sum_{j=2}^{N} \frac{1}{j} \sim \log N$. This finishes the proof.

Question 4 Uniform Convergence over Half-spaces (16 Points)

Let

$$\mathcal{F} = \left\{ \mathbf{1}\{a^\top x + b \le 0\} : a \in \mathbb{R}^2,\ b \in \mathbb{R} \right\}$$

be the class of indicators of half-spaces in $\mathbb{R}^2$.

(a) Prove that $\mathrm{VCdim}(\mathcal{F}) = 3$. [6 points]

Recall: $\mathrm{VCdim}(\mathcal{F}) = k \iff$ Some set of k-points can be shattered by $\mathcal{F}$, and no $(k+1)$-points can be shattered by $\mathcal{F}$. For this problem, it might be useful to consider two cases: (i) one point lies in the the convex hull of the other three points, (ii) all four points are in convex positions.

(b) Let $X_1, X_2, \ldots$ be i.i.d. observations from an arbitrary distribution P on $\mathbb{R}^2$, and define

$$P_n f = \frac{1}{n} \sum_{i=1}^{n} f(X_i), \qquad Pf = \mathbb{E}[f(X_1)].$$

Prove that there exists a finite constant $C > 0$ such that

$$\mathbb{P}\left(\sup_{f \in \mathcal{F}} |P_n f - Pf| \le C \sqrt{\frac{\log n}{n}} \text{ for all sufficiently large } n \right) = 1.$$

Using the above, conclude that $\mathcal{F}$ is a Glivenko–Cantelli class:

$$\sup_{f \in \mathcal{F}} |P_n f - Pf| \longrightarrow 0 \qquad \text{almost surely.}$$

[10 points]

Hint: Note that $\mathcal{F}$ is a bounded class of functions, and hence, has bounded difference property.

Solution:

(a) Choose three noncollinear points x_1, x_2, x_3. Each singleton can be separated from the other two by a line, and taking complements realizes each two-point subset. The empty set and the full set are also realized. Hence, these three points are shattered, so

$$\mathrm{VCdim}(\mathcal{F}) \ge 3.$$

Now consider any four points.

If one point, say x_4, lies in $\mathrm{conv}\{x_1, x_2, x_3\}$, then every half-space containing x_1, x_2, x_3 must also contain x_4. Thus, the labeling

$$(1, 1, 1, 0)$$

cannot be realized.

Otherwise, the four points are in convex position. Label them cyclically as x_1, x_2, x_3, x_4. The diagonals $[x_1, x_3]$ and $[x_2, x_4]$ intersect. If a half-space contained x_1, x_3 and excluded x_2, x_4, then their intersection point would belong both to the half-space and to its convex complement, which is impossible. Hence, the alternating labeling

$$(1, 0, 1, 0)$$

cannot be realized.

Therefore, no four points can be shattered, and

$$\boxed{\mathrm{VCdim}(\mathcal{F}) = 3.}$$

(b) Define

$$Z_n = \sup_{f \in \mathcal{F}} |P_n f - P f| .$$

Since $\mathcal{F}$ is a $\{0,1\}$-valued VC class of dimension 3, the standard VC–Rademacher bound gives, for some universal constant $C_0 < \infty$,

$$\mathbb{E}[Z_n] \le C_0 \sqrt{\frac{\log n}{n}}.$$

If one observation X_i is replaced by X_i', then

$$\sup_{f \in \mathcal{F}} \left| P_n f(X_1, \dots, X_i, \dots, X_n) - P_n f(X_1, \dots, X_i', \dots, X_n) \right| \le \frac{1}{n}.$$

Consequently, Z_n also changes by at most $1/n$. McDiarmid's inequality therefore yields

$$\mathbb{P}\left(Z_n - \mathbb{E}[Z_n] \ge t\right) \le \exp(-2nt^2).$$

Take

$$t_n = \sqrt{\frac{\log n}{n}}.$$

Then

$$\mathbb{P}\left(Z_n - \mathbb{E}[Z_n] \ge t_n\right) \le n^{-2},$$

and hence

$$\sum_{n=1}^{\infty} \mathbb{P}\left(Z_n - \mathbb{E}[Z_n] \ge t_n\right) < \infty.$$

By the Borel–Cantelli lemma, almost surely, for all sufficiently large n,

$$Z_n \le \mathbb{E}[Z_n] + t_n \le (C_0 + 1) \sqrt{\frac{\log n}{n}}.$$

Thus, with $C = C_0 + 1$,

$$\mathbb{P}\left(Z_n \le C \sqrt{\frac{\log n}{n}} \text{ for all sufficiently large } n \right) = 1.$$

Finally, since $\sqrt{\log n / n} \to 0$,

$$\boxed{\sup_{f \in \mathcal{F}} |P_n f - P f| \longrightarrow 0 \qquad \text{almost surely.}}$$

Therefore, $\mathcal{F}$ is a Glivenko–Cantelli class.